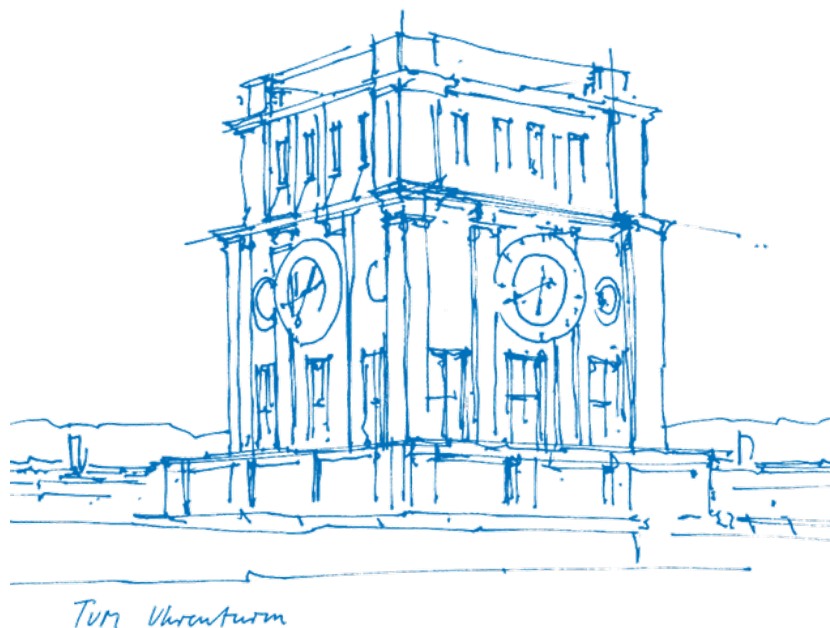


Master's Thesis in Information Systems

Xin Wen

Document-Based Process Modeler



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Document-Based Process Modeler

Dokumentatenbasierte Bearbeitung und Änderung von
Prozessmodellen

Thesis for the Attainment of the Degree
Master of Science

at the TUM School of Computation, Information and Technology,
Department of Computer Science,
Chair of Information Systems and Business Process Management (i17)

Examiner

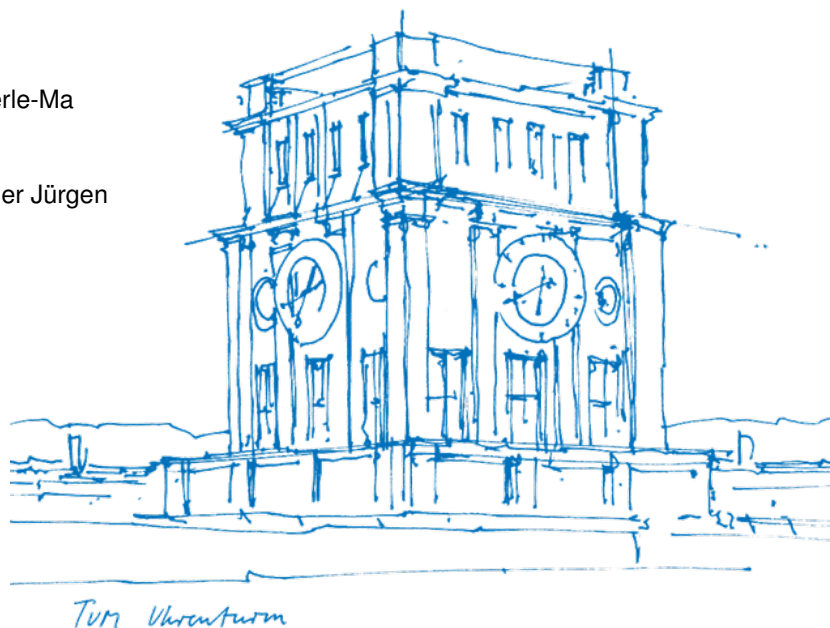
Prof. Dr. Stefanie Rinderle-Ma

Supervised by

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Submitted on

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Xin Wen

Abstract

Business process is valuable asset of an organization. But most of them are store in unstructured format, e.g., Word documents, PDF files. Manual translation from unstructured documents to structured process models is time-consuming and error-prone. LLM has shown as a great technique to support creating proces modeling efficiently, but it still cannot fully replace human in process modeling. Hybrid human-LLM process modeling is a promising way to leverage the strength of both human and LLM, and mitigate their weaknesses. However, existing tools and approaches still have limitations in terms of usability, integration, and functionality, especially in the use case for creating process models from unstructured documents. This work aims to address these limitations by proposing and developing an integrated system that supports process modeling from documents using LLMs with a more seamless, efficient and user-friendly workflow. **Keywords:** *Process Modeling*

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Introduction

Motivation

Business process is a valuable asset of an organization. But most of them are stored in unstructured format, e.g., Word documents, PDF files. Manual transformation from unstructured text to structured process models is time-consuming and error-prone.

In the era of Gen-AI, LLM has proved to be a promising solution to support creating process modeling efficiently. However, on one hand it still cannot fully replace human in process modeling especially when domain knowledge is needed. On the other hand, existing LLM-based process modeling tools and approaches still have limitations especially in the scenario for creating process models from documents. For example, the typical "one view one model" approach make it hard to manage for large systems, where multiple process and subprocess models are needed. In addition, the usual separate interface of the document and the generated model make it hard to verify the result. Besides, user need to manually extract relevant information from documents and input it into the conversational chatbot.

To address these limitations, this work proposes an integrated tool that supports process modeling from documents with a more seamless, efficient, and user-friendly workflow by offering the following core features: (1) manage all the related documents and models for one use case within one unified view (2) allowing users to upload documents and select passages from the documents to serve as input for LLM-based process model generation directly, (3) providing a clear visual link between the text and the generated model through display the document and the model side by side

Research Questions

This work addresses the following research questions:

RQ1 How do existing tools supported AI assisted process modeling with multiple documents as input?

This question investigates how existing approaches and tools support process modeling from documents by identifying their functionalities and capabilities. It helps guide the design of the proposed system.

RQ2 Which layout and properties are beneficial for an integrated system that supports AI assisted process modeling with multiple documents as input?

This question addresses the design of the proposed system. It focuses on what functionalities should be integrated and how to integrate them, aiming to enable a seamless modeling workflow with high usability.

RQ3 How do user perceive the benefits and limitation of the system?

This question evaluates the proposed system from the users' perspective. It aims at understanding how users perceive the system's benefits and limitations. Its results provide insights into the system's strengths and weaknesses and help identify potential design issues and areas for improvement.

Contribution

The contribution of this thesis consists of the following aspects:

Firstly, a systematic review of the existing tools and approaches for AI-assisted process modeling is conducted, which provides insights into the current state of the art and guides the design of the proposed system.

Secondly, an integrated system is designed and developed to support process modeling from documents using LLMs. The system offers a more seamless, efficient, and user-friendly workflow by integrating multiple functionalities, such as document management, model generation, and traceability between text and model.

Thirdly, the proposed system is evaluated through a user study with process modeling experts. The evaluation results provide insights into the system's benefits and limitations from the users' perspective, which can inform future improvements and further research in this area.

Methodology

The process of this thesis employs the research cycles of DSR from Hevner[1]: The relevance cycle: the first cycle is done through the communication with modeling experts and proposes the tool and elicits the requirements in section 3.3 for the system.

The rigor cycle: it is done through the review in section 2. The design cycle: the first cycle is done through the delivered solution of RQ2 and evaluation result of RQ3. Through the analysis and discussion of the evaluation result, we can identify the strengths and weaknesses of the proposed tool, and further improve the design of the tool in the next iteration.

Evaluation

The evaluation of the proposed system is done with 4 experts in qualitative way. It consisted of task-based activities and a follow-up questionnaire to collect user feedback on the tool's usability and perceived usefulness.

Structure

The remainder of this thesis is structured as follows:

Chapter 2 reviews the related work of LLM-based process modeling tools. Chapter 3 presents the design solution of the tool, including its requirements, architecture, and functionalities. Chapter 4 describes its technical implementation. Chapter 5 describes the evaluation of the proposed system, including the methodology, results. Chapter 6 discusses the implications of the findings and potential future work. Finally, Chapter 7 summarizes the current work and outlines future work.

Related Work

Foundational Concepts and Technologies

In this section, we briefly introduce the foundational concepts and technologies relevant to this thesis.

Business Process Modeling (PM) BPMN, LLM, BPM, RPST,

While many tools exist that deal with analysis of large document sets, to the best of our knowledge no integrated tool with the dedicated purpose of creating Process Models from documents set exists. Therefore here we will explore some related tools.

In total, eleven AI-assist modeling tools are explored during the literature review (See table 2). Here, we mainly focus on the key approaches and the distinguishing features they offer.

Table 1

Literature review: horizontal research

<i>Keywords for search</i>	<i>hits</i>	<i>selected</i>	<i>Selection criteria</i>
allintitle: process modeling generative AI	12	1	Process modeling tool
allintitle: process modeling LLM[s]	16+7	5+0	Process modeling tool
allintitle: process modeling large language	49	1	
allintitle: conversational process modeling	8	1	Process modeling tool
allintitle: BPMN generative AI	2	0	
allintitle: BPMN LLM[s]	12+8	2+1	
allintitle: BPMN large language	7	1	
<i>Results horizontal search</i>	121	11	
<i>Results vertical/backward search</i>			
Added papers		1	
<i>Overall:</i>		1	

Source: <https://scholar.google.com/>

In the 12 hits of the first search query("allintitle: process modeling generative AI"), the tool ProMoAI [2] is identified. It is the first tool shown in the academic literature to leverage LLMs for automated process model generation, introduced in 2024, and is frequently referenced in subsequent work. ProMoAI enables the transformation of natural language into process models and supports iterative refinement through feedback loops, demonstrating the feasibility of LLM-based process modeling.

In the second query ("allintitle: process modeling LLM"), among 16 hits, 5 tools are identified. The first two noticed is BPMN-Chatbot [3] and BPMN Assist [4]. BPMN-Chatbot is an early one in 2024 with key strength in token efficiency[5], compared with ProMoAI's approach. Its method was introduced in. BPMN Assist [4] is a very latest one. BPMN Assistant [4] represents a more recent development, emphasizing structured intermediate representations (e.g., JSON) to improve reliability, editing capabilities, and performance in interactive modeling scenarios.

The tool KICoPro is introduced in [6], focusing on human-in-the-loop process modeling, where users collaborate with AI systems during model creation.

The tool HyperMode is introduced in [7], which combines LLM-based generation with a direct manipulation interface, allowing users to refine process models interactively by selecting and modifying elements within the diagram.

Additionally, the tool BPMNGen is firstly introduced in [8] then further developed and evaluated in [9]. It is a different one using NLP, with empirical evaluations showing strong semantic quality for simple to moderately complex processes.

In the last search attempt for term "process modeling" by combining it with "large language", one more tool PRODIGY is explored mentioned within the paper [10]

In the search query ("allintitle: conversational process modeling"), the only tool AutoBPMN.AI [11] is found.

Finally, combining “BPMN” with "generative AI", "LLM[s]", or "large language" led to the identification of 3 additional tools, including BPMN-Chatbot++ [12], LLM4BPMNGen [13], and Nala2BPMN [14].

BPMN-Chatbot++[12] is found in Tool no name [15]

LLM4BPMNGen[13]

Nala2BPMN [14]

In summary, although a growing number of tools leverage LLMs to support process modeling, none provide a fully integrated solution for automatically deriving complete process models from large document collections.

In summary, while numerous tools leverage LLMs for generating process models from natural language, they predominantly focus on single textual inputs and interactive refinement. There is still no integrated solution specifically designed to process and synthesize large document sets into coherent, unified process models. This highlights a clear research gap and motivates the need for approaches that bridge document analysis and process model generation in a scalable and integrated manner.

Table 2
Related work on state-of-the-art AI-assisted process modeling tools

Tool	Year	Input Form	Output Notation	Key Contribution / Difference
ProMoAI [2]	2024	Natural language text + iterative feedback	BPMN / PNML	Generates process models via constrained Python code and intermediate POWL representation; strong focus on correctness and error handling.
BPMN-Chatbot [3], [5]	2024	Text or voice input + interactive refinement	BPMN	Conversational web-based tool with strong token efficiency and correctness; supports iterative modeling and versioning.
Nala2BPMN [14]	2024	Natural language text	BPMN	Hybrid pipeline combining LLM extraction with algorithmic BPMN construction; handles ambiguity and incomplete input.
PRODIGY [10]	2024	Natural language + enterprise documents (RAG)	BPMN Sketch Miner textual syntax	Enterprise-oriented tool leveraging retrieval-augmented generation for process modeling from organizational knowledge.
LLM4BPMNGen [13]	2025	Text, audio upload, or voice input	BPMN	Multimodal input support with structured prompting; improves coverage of BPMN elements and generation robustness.
BPMN-Chatbot++ [12]	2025	Text/voice (enhanced interaction)	BPMN	Extension of BPMN-Chatbot with improved prompting, interaction quality, and modeling performance (focus on efficiency and usability).
AutoBPMN.AI [11]	2025	Natural language text	CPEE-Tree	Output in executable format
HyperMode [7]	2025	Natural language + direct manipulation (UI interaction)	BPMN diagrams	Combines LLM-based generation with direct manipulation interface, enabling users to refine models via selection-based interaction and prompts.
BPMNGen [8], [9]	2026	Natural language text + conversational refinement	BPMN	LLM-based conversational framework with strong focus on human-centered evaluation (semantic quality, cognitive load, usability).
BPMN Assist [4]	2026	Natural language text (editing + creation)	BPMN	Introduces structured JSON intermediate representation for reliable and efficient editing; improves latency, token efficiency, and editing success.
KICoPro [6]	2026	Natural language + human-in-the-loop interaction	BPMN	Emphasizes human-AI collaboration and interactive refinement in process modeling (focus on usability and co-creation).

AutoBPMN.AI

Here, we provide a more detailed in depth review of the AutoBPMN.AI tool, as we will leverage its LLM service for the automated process model generation in our proposed system.

AutoBPMN.AI provides an LLM-backed service that turns natural language process descriptions into an *executable* process representation (CPEE-Tree) [11]. The core interaction is conversational: the user describes a procedure in free-form text, and the service produces (and maintains) a structured model corresponding to that description.

Model creation is performed iteratively over multiple turns. After the initial model draft is generated, the user can issue follow-up instructions (e.g., clarifications, corrections, additions, or restructuring requests). The service uses the conversational context to interpret each instruction and updates the executable model accordingly, enabling step-by-step refinement rather than a single one-shot generation.

In summary, the AutoBPMN.AI LLM service operates as a dialogue-driven modeling loop with two key artifacts exchanged between user and system: natural language messages as input and an updated CPEE-Tree model as output at each iteration.

In this thesis, we will leverage the LLM REST service provided by AutoBPMN.AI for model generation and refinement within our proposed system.

Solution Design

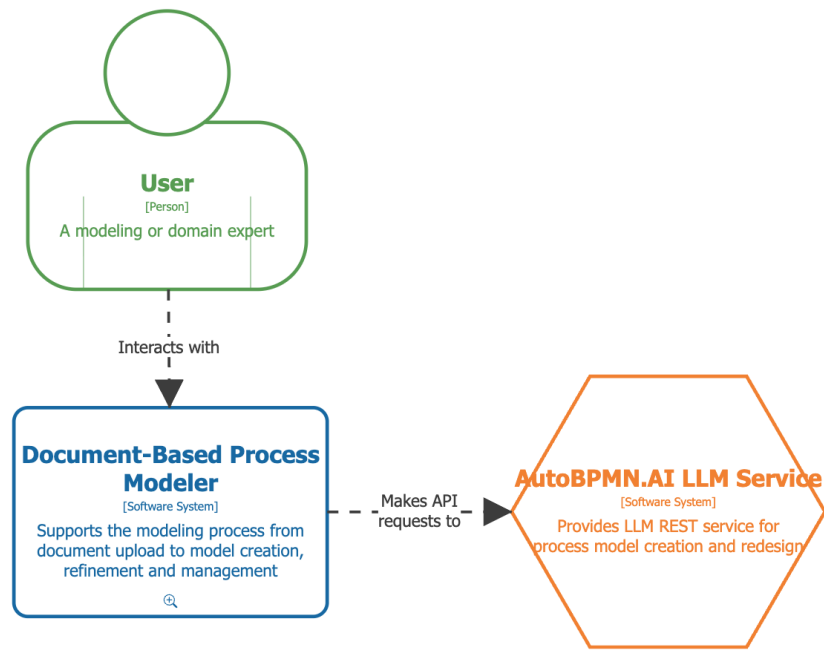
Requirements

Context and Goal

First, we define the context and goal of the solution design. The system is provided as a web-based application, as illustrated in Figure 1. The target users include both process modeling experts and domain experts who need to create process models based on documents. To improve modeling efficiency and support non-expert users, the system leverages the LLM service of AutoBPMN.AI for automated model generation. The overall goal is to provide a single, integrated platform in

which users can complete the entire modeling process from document upload to model creation, modification and subsequent model management.

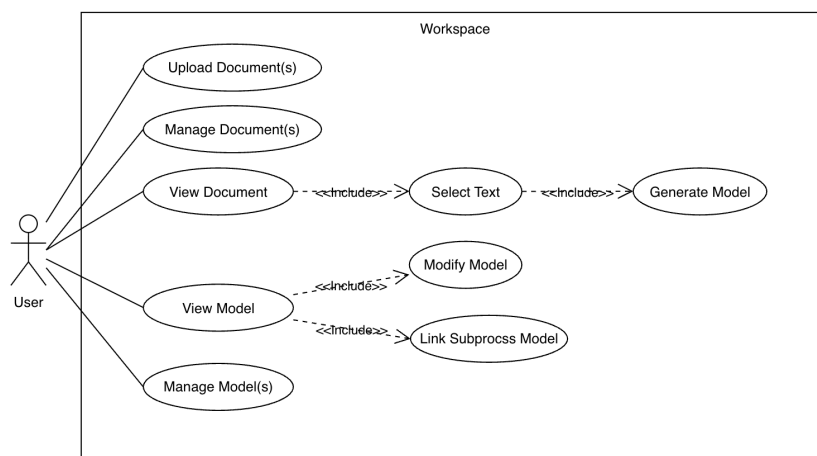
Figure 1
System context



Requirements Elicitation

When starting the work the following requirements for the system were defined.

Figure 2
Usage model



Requirements Elicitation

Functional Requirements

Functional requirements define the specific behaviors, actions, and features a system must perform. They outline input/output processing, data manipulation, and user interactions necessary for the system to meet its objectives. These requirements are essential for to build a prototype for this thesis, ensuring system functionality and user needs are met. [LINK TO WIKIPEDA](#)

It shall allow users to select multiple passages from one document to serve as input for LLM-based process model auto-generation.

It shall provide a clear visual link between the text and the generated model.

It shall embed the source text within the generated process model for traceability purposes.

It shall allow users to establish process-subprocess relationships between models.

It shall record user behavior and interactions with the system for analysis purposes.

One additional requirement is that the system shall be able to support document update.

- *Upload Documents.* The user uploads multiple documents in textual format, e.g., TXT, DOCX, PDF.
- *Select Text Passages.* The user selects multiple passages from one uploaded document, then clicks the Generate button to .

Design Constraints

- It shall be implemented as a web-based application.
- The frontend shall be developed using native web technologies (JavaScript, HTML, and CSS).
- *Process Model Visualization.* The visualization of the process model shall be implemented using the CPEE Layout¹ library.
- *LLM Service Integrati.* The LLM-backed process model generation shall be implemented using the API from AutoBPMN.AI.

¹<https://github.com/etm/cpee-layout>

User Interface and Interaction Design

The design overview is described from the user perspective.

Firstly, we define the following concepts to better describe the system:

- Project: a use case consisting of one or more documents for which process models needs to be created.
- Workspace: the unified page/interface where the modeling workflow is performed, from document upload to model export.
- Traceability: the ability to navigate between source text and the corresponding process model.
- Project graph: a network-style visualization of relationships among models and documents within a project.

The system consists of three main pages: a home page for managing projects, a workspace page where the modeling workflow takes place, and a statistics page for viewing project-related metrics.

System Views

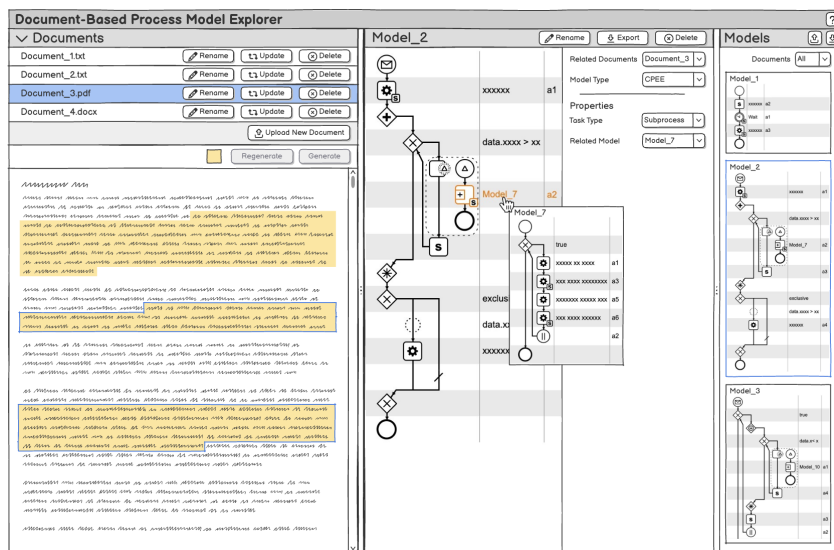
The system is designed to contain three pages: a home page where a user can create a new project or open an existing project, a workspace page where the modeling workflow takes place, and a statistics page where the user can view the statistics of the project.

System Views

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Workspace. The layout design of the user interface is shown in the figure 3 in a 3-pane layout. The left pane is for document uploading and viewing, the middle pane is for process modeling and the right pane is for process models viewing and project graph visualization. The details are divided as the following: a) Document List b) Document Viewer c) Model Editor d) Model List e) Project Graph.

Figure 3
Workspace UI Layout



The user interface layout of the workspace page.

Modeling Workflow Design

As the core part of the system, this section describes the modeling workflow that happens in the workspace page in Use Case Diagram. Figure: .

Design Difficulties

Core Functionalities Design

Document Management

Document Update

The document update is planned and designed to

Document Interaction

text selection text selection editing

Process modeling

Human-in-the-loop Process Modeling and Version Control.

Use Case: Model Generation. By click the button an advance mode is desgin

Use Case: Model Modification. a. Manual edit b. Update selected text and apply the changes. c. Refinement by sending prompt to LLM:

Input: prompt only and the current Model

d. Regeneration:

Input: type a. current model + changed selected text passage b. advanced mode: prompt + current model + changed selected text passage

Use Case: Establishing Process-Subprocess Relationships. - a review step for each LLM-generated model.

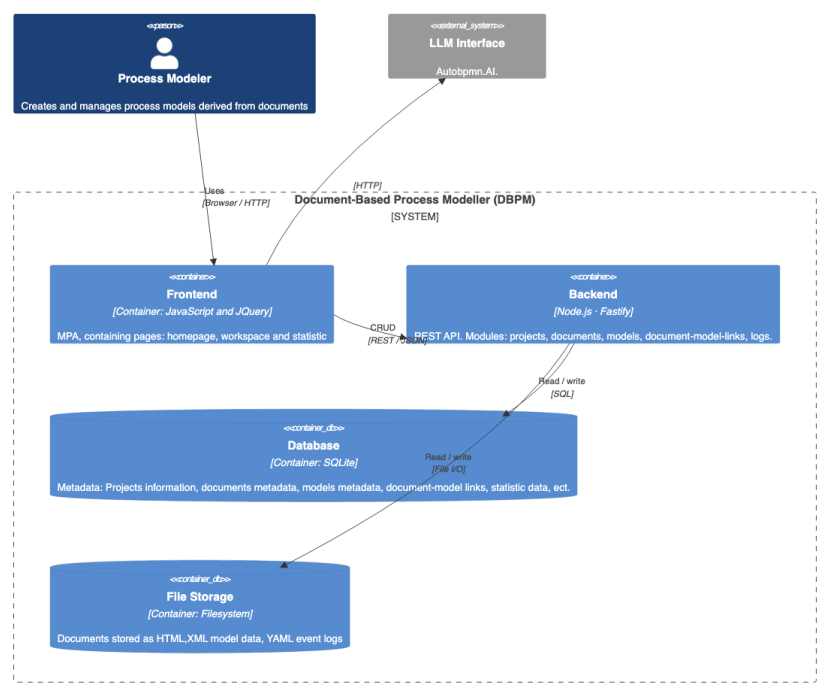
Augmented Model Data.

Interactive Design Difficulty

System Architecture

The system architecture is shown in the figure 4. The system follows the common web-application client-server pattern.

Figure 4
System Architecture

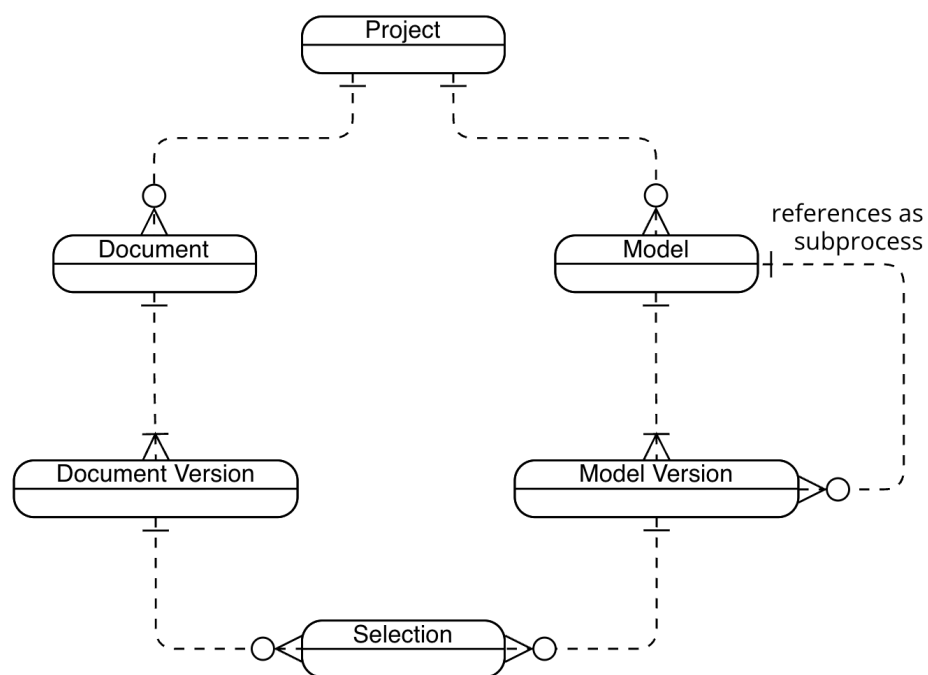


The system architecture of the document-based process modeling tool.

Data Model

Figure 5

The conceptual data model of the system.



Implementation

[16].

Frontend

Architecture

Figure 6

Frontend codebase structure

```
frontend/
├── api/
│   ├── base.js
│   ├── documents.api.js
│   ├── models.api.js
│   ├── projects.api.js
│   ├── document-model-links.api.js
│   └── ...api.js
├── modules/
│   ├── document/
│   │   ├── file.js
│   │   └── selection.js
│   └── model/
│       ├── wfadaptor.js
│       ├── endpoints/
│       ├── themes/
│       └── utils/
├── pages/
│   ├── home/
│   │   ├── init.js
│   │   └── ui/
│   │       ├── projects.ui.js
│   │       ├── project-creation.ui.js
│   │       └── dashboard.ui.js
│   ├── stats/
│   │   ├── init.js
│   │   ├── store.js
│   │   └── ui/
│   └── workspace/
│       ├── init.js
│       ├── services/
│       │   └── workspace.service.js
```

State Management

pub-sub

Figure 7
store implementation

```
export class Store {
  constructor(initialState = {}) {
    this.state = { ...initialState };
    this.subscribers = new Set();
  }

  subscribe(fn) {
    this.subscribers.add(fn);
    return () => this.subscribers.delete(fn);
  }

  notify(patch) {
    this.subscribers.forEach((fn) => fn(patch));
  }
}
```

Document Processing

Text Selection Rendering

Virtual Overlay Strategy.

Virtual Overlay Strategy.

Modeling***Rendering Customization******Architecture*****Backend*****Architecture***

The tech-stack for the backend is currently chosen as Node.js with Fastify, as it provides out-of-the-box support for schema validation. As shown in the figure 8, the example code snippet of the backend

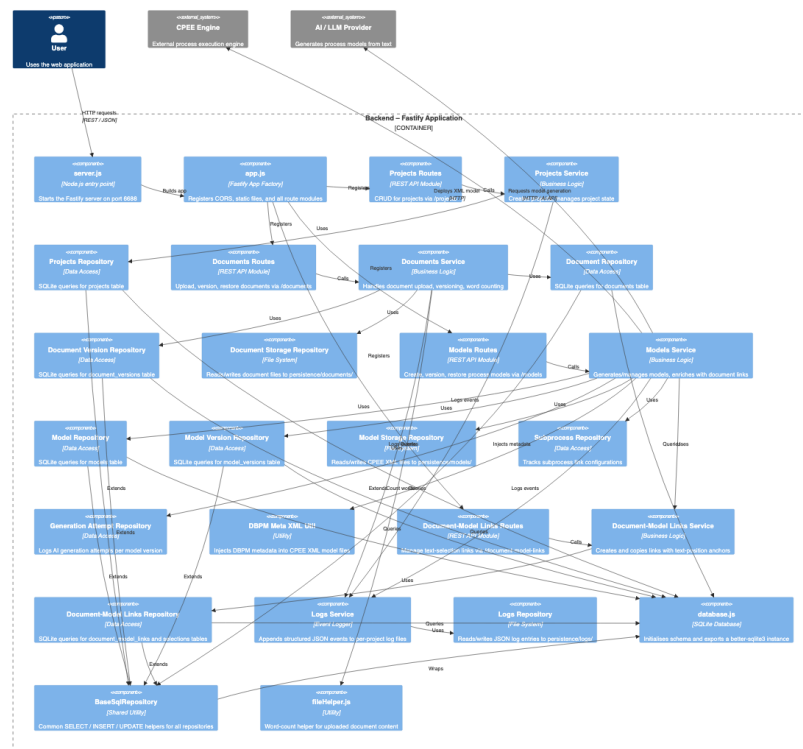
Figure 8

Example JSON schema used in the backend for text quotes

```
const textQuoteSchema = {
  type: "object",
  required: ["exact"],
  additionalProperties: false,
  properties: {
    exact: { type: "string" },
    prefix: { type: "string" },
    suffix: { type: "string" },
  },
};
```

The system uses a TextQuoteSelector for robust anchoring, as defined in the W3C Web Annotation Model (Text Quote Selector section) The backend implementation follows the layered Controller-Service-Repository architecture.

Figure 9
Backend Component



The backend component of the document-based process modeling tool.

Database Schema

Firstly, Project, Document, Model and Document Model Link are the minimal tables needed to support the proposed system. It is also how the system was implemented when it first adopted the database. The Document and Model tables were used to store metadata like name, words count of a document. Theoretically, the Document Model Link table can be stored as an attribute within the model version, since the system is currently designed to allow one model to be derived from only one document. But it is a decision made very early to store it in a separate table to easily support a potentially many-to-many (M2M) relationship in the future.

The Model Version table is added when the model versioning feature is about to be implemented. The Document Version table is added at the same time to prepare for supporting the Document Update feature, as both tables follow a similar logic at the DB operation level.

The Subprocess Link table is an additional table used to support the Project Graph. As this relationship information has been embedded in Model Data, the model visualization itself does not need this data. However, for the project graph, the relationship is hard to be parsed from the model data; therefore, such a table is designed to easily retrieve the process-subprocess relationship.

The Selection and Selection History tables are then added when designing the Document Update feature. Before the selection data (e.g., color, position) for a Document Model Link is stored as an attribute in the format of a stringified JSON object. To better support auto-reanchoring for document upload, Selection and Selection History are designed as separate tables. The separate Selection table can make selection data easier to retrieve for reanchor operations. Besides, a Selection History is designed to make it easier to track the quality of reanchor.

Finally, the Generation Attempt and Version Event tables are the two tables designed to monitor the system behavior. The Generation Attempt table is used to monitor the accept/decline decision for the LLM-generated output. The Version Event is mainly to store the manual edit behavior or a system auto-update behavior, e.g., selection reanchor for document update.

Evaluation

Evaluation Procedure

The developed system was evaluated with four modeling experts to assess its usability and perceived usefulness, as well as to gather feedback on potential improvements and additional features. The evaluation process included a short demonstration, task-based activities, and a questionnaire. Each evaluation session lasted approximately 40 minutes.

The session was structured as follows: It began with a demonstration to introduce the tool's end-to-end workflow and key features. After that, the participants were asked to complete two tasks. The first was an exploratory task based on a fictional B2B order-processing scenario, consisting of two wiki-style documents in DOCX format. It was designed to help participants familiarize themselves with the system. The second was a more realistic process modeling task based on an open-source SOP document² in PDF format. For both tasks, participants were not given strict instructions on

²DNRC SOP Document

how to proceed, allowing them to explore the system freely. Due to the known issue of the PDF file rendering, the link to the online version was provided. If time permitted, participants could additionally use the system with their own documents.

Following task completion, participants completed a questionnaire including the following open-ended questions:

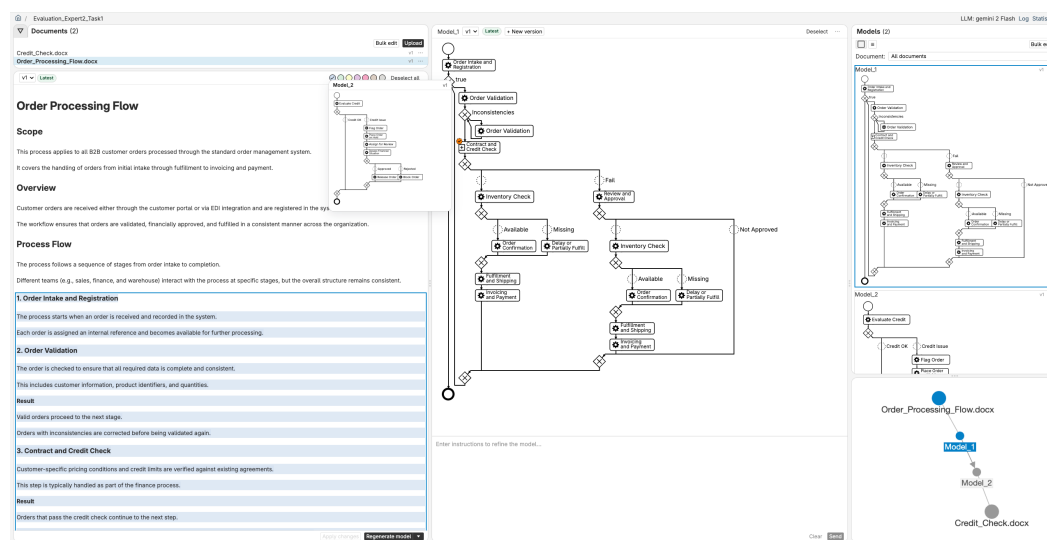
- (1) What aspects of the tool did you like while using it?
- (2) What aspects of the tool did you not like or find problematic?
- (3) What improvements or additional features would you most like to see in the tool?
- (4) If you needed to model processes from documents in the future, would you consider using this tool? Why or why not?

Evaluation Results

Overall Perception and Task Completion

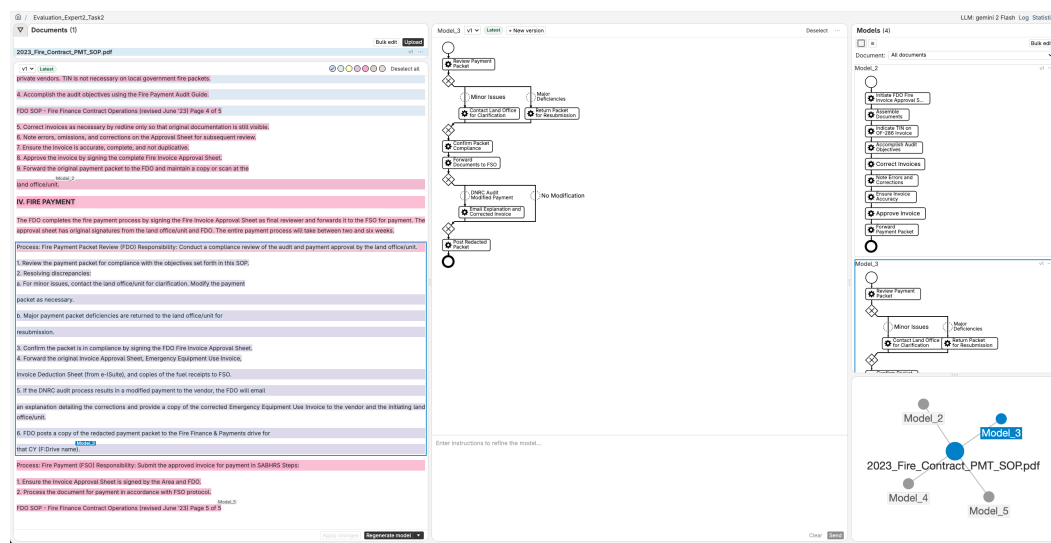
Overall, participants perceived the tool as usable and intuitive. Figure 10 and Figure 11 show the modeling results from one expert for the first and second tasks, respectively.

Figure 10
Modeling result



Modeling result from an expert for the first task.

Figure 11
Modeling result



Modeling result from an expert for the second task.

One interesting observation is that while most participants tended to start with, or at least try once, selecting larger portions or all of the text when reading a new document, one participant began by creating models from smaller segments. These different strategies are illustrated in Figure 12.

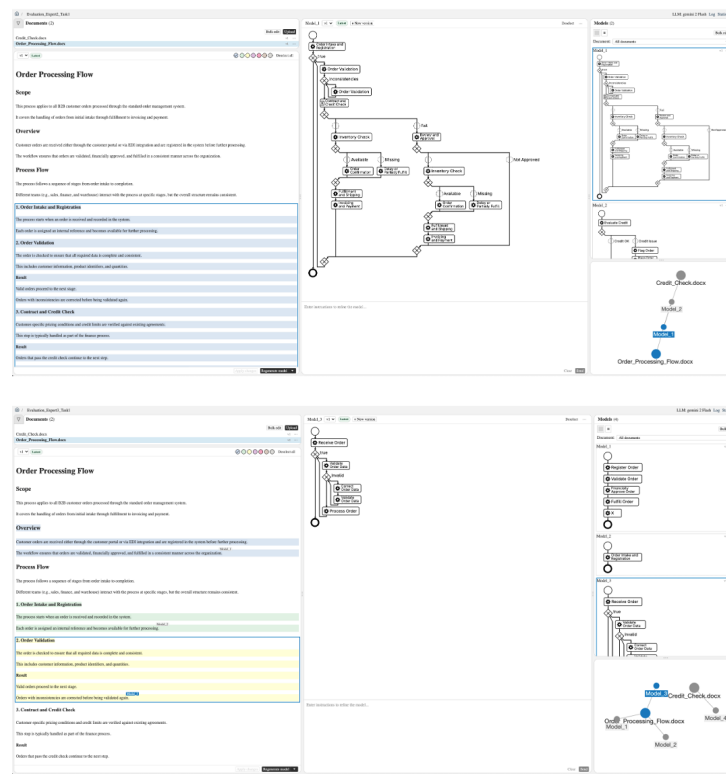
For detailed task completion and feedback, the participants showed varied interests and offered differing perspectives on the system. The result on each feature is summarized in Table 3, and will be described in detail in the following sections.

Positive Aspects

Firstly, the following aspects are generally perceived positively by the participants: the performance of the integrated LLM service, the easy-to-establish process-subprocess relationship, and model version control.

Furthermore, several additional positive aspects were mentioned. First, the embedding of relevant information, such as selected text and prompt history, within the model’s XML data is considered a well-integrated design feature. Secondly, logging and statistical information elicited interest among some participants due to its potential for further analysis. Thirdly, the project graph visualization was mentioned as helpful, especially for larger systems.

Figure 12
Different modeling strategies



Different modeling strategies observed during task execution (above: larger part selection; below: step-by-step selection)

Problems in Interaction and Use

The following outlines the main problems and challenges reported by participants, as well as notices observed during the evaluation.

First, for the document's rendering and interaction, only one feedback regarding the readability issue of PDF documents was mentioned in feedback. But it is observed that one participant started with reading the original PDF online firstly and one participant tried to use browser-extension to search for keywords.

Regarding the modeling process design, two problems were mentioned. First, the step to start a new modeling by a explicit deselection step of the current model was mentioned. Two reply received in the feedback, but it is a feature generally perceived as not intuitive immediately and needs additional clarification. Second, due to the sufficient good quality of the initial model, the review step was

Table 3
Summary of Evaluation Results by Functionality

Functionality	Positive Aspects		Problems and Limitations	
	Item	n	Item	n
Document Rendering and Interaction			Readability issue	1
			Expected AI-assisted search capability	1
Modeling Process			Unintuitive to start a new modeling by deselecting the current model	2
			Unnecessary review step (of the initial generated model)	2
Integrated LLM Service	Fast	2	Lack of transparency in the conversational modeling logic	2
	Good quality	2		
Model Versioning	Helpful	3	Expected version comparison	1
Augmented Model Data	Well-considered	2		
Text-Model Link	Clear	1	Expected fine-grained traceability	2
Process-Subprocess Relationship	Convenient	3		
Project Graph	Helpful for larger systems	1	Node types not clear	1
Log and Statistics	Potentially useful for analysis	1		

n = number of participants who mentioned each item

mentioned by one participant as totally unnecessary as there is version control and by another one as the initial generated model was mentioned as an unnecessary step.

Another main problem mentioned and observed is that the system's conventional modeling logic is not transparent to users and needs additional explanation. While the initial model generation from text was relatively clear, the model modification process was less transparent. They did not understand how their inputs were processed or how the system used the selected text and existing model.

Potential Improvements and Additional Features

Besides the problems found during system use, many valuable insights were collected.

For the existing features, improvements to the project graph visualization were mentioned most often. This includes clearer differentiation and identification of elements (i.e., what type of element a node is and which version it represents), as well as using more expressive visual designs, such as size, color saturation, and distance, to show additional information beyond type.

For document-related features, an AI-assisted document search and interaction function was suggested. This would allow users to query the document by providing an instruction, after which the retrieved parts can be automatically selected and used directly or as a basis for further adjustment in model generation.

For the process modeling features, one notable desired improvement emphasized by the participants is the need for finer-grained traceability between source text and model. When selecting larger text passages, only a subpart of the text is actually used and reflected in the generated model, and it is unclear which parts are actually used. Therefore, a clearer indication of the actual text used for model generation is required, ideally at the level of individual tasks. Secondly, model comparison was suggested as a valuable feature to help users understand and verify changes between different model versions, making it easier to identify modifications and track model iterations over time. Beyond this, participants also expected more general advanced AI modeling capabilities, including explainability of the generated model through conversational interaction with chatbots, as well as support for model quality evaluation and automatic improvement.

Future Use Intention

Three participants indicated an intention to use the system in scenarios involving the creation of multiple models. In such cases, the system's features, such as text-model link, process-subprocess relationship, and model management functionalities, were considered helpful. One participant additionally raised the concern that automatically generated models may bypass the user's need to fully understand the underlying document, potentially introducing bias. Hence, the fine-grained traceability, as suggested by the participant, could help mitigate this problem. Another participant again pointed out the need for the AI-assisted document search capability as suggested.

Discussion

This chapter discusses the evaluation's findings and reflects on their implications for system design.

Usability and User Experience

Regarding the usability issues. While some features can be mitigated by design, some are hard. As this tool eliciate a new way of process modeling, it requires a fam to let user get used to the new way of modeling.

User Expectations

1. Document -
2. Modeling The User show more expectation than, although as we are aming to keep a transparent traceability between . User show higher expectation than the normal tool according to the feedback. If we want the . For a such tool, if the desgin goal is to let them

Limitations

The materail we used in the evaluation is Although, there is a real world document. The participants are not using their own documents to solve a real task from their own work area. Besides, the

Conclusion

Summary

In this work, we propose and develop a process modeling tool, especially for the needs process modeling from documents. We explore two possibilties

Here we summarize the research questions:

RQ1 How do existing tools supported AI-assisted process modeling with documents?

As elaborated in Chapter02, the current tools in research / academic area, mainly focus on the quality of the generated model, accoding to the provided description. To our best knowledge, there is no

tools right now to support model auto-generation from document, therefore, to do so an export has to copy and paste text themselves

RQ2 How can an integrated system be designed to support process modeling from documents? RQ2 Which layout and properties are beneficial for an integrated system that supports AI assisted process modeling with multiple documents as input? - ?

RQ3 How do user perceive the benefits and limitation of the system? RQ3 What are advantages and disadvantages of such a system?

Perceived benefits: Intuitive and reduce the tool switching time..., versioning Perceived limitations: More features is expected

Participants found the system particularly valuable for creating and exploring multiple models from documents using features like text-model link and process-subprocess model link. However, for simpler tasks, such as building a model from a single document, the system offers limited advantages compared to existing tools.

In summary, we extend the functionality of LLM-based modeling processing to in two area: 1. we offered a new input modality for LLM-based modeling - the input text is selected from the document. 2. a traceability. 3.

Future Work

Future work includes: - Improve and Extend the current features. e.g. the rendering issue , improve ..., text search,,, or support more documents type by leveraging other types... For such a system a lot of stuff can be done. - Integrated some document intelligent from simple e.g. using AI-to summarize the content, to advanced like chatbot query - Fine-tuning the LLM for more precise text extraction, as discussed, since one of the key features is the traceablity, a dedicated for such case can be offered.

I. Research Implications

II. Future Research Directions

Future research can be conducted in depth across four areas: user interaction, system functionality, intelligent assistance, and scenario implementation. First, optimize user-intuitive interaction and visualization design, simplify operational steps, and improve multi-format document rendering capabilities. Clearly distinguish model elements and version relationships through visual cues to enhance overall usability. Second, establish a refined model management and traceability system. Develop features such as version comparison, fine-grained text-to-model traceability, and intelligent document information filtering to enhance full-lifecycle modeling support capabilities. Third, research explainable AI-assisted modeling mechanisms. Clarify AI input rules and reasoning processes to enable predictable modification outcomes and controllable workflows, ensuring user autonomy while advancing intelligence levels. Fourth, integrate search-enhanced generation technology with conversational interaction to build cross-document contextual modeling capabilities. Develop automated model quality assessment and optimization functions to further improve modeling efficiency and quality. Fifth, conduct adaptation research for practical scenarios such as complex business processes and multi-document collaboration. Promote the integration of the tool with existing systems to establish standardized application paradigms, thereby enhancing the tool's engineering implementation capabilities and expanding its industry applicability.

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Appendix